



80%	91%	IET	Land IPT
FIRST CLASS BENG	VAIDAR CAPSTONE	PRIZE WINNER 2026	TEST CONTACT

Accomplished **Hardware Test Engineer & FPGA Verification Specialist** with a First-Class 80% BEng (Hons) in Electronic and Computer Engineering and recipient of **The IET Prize 2026**. Technical expertise spanning military-grade electro-optical surveillance director qualification, automated hardware-in-the-loop (HIL) testbench design, and automated digital logic verification. Developer of **VAIDAR**, a novel Python-native execution engine bridging machine learning models with physical AMD Xilinx Artix-7 FPGA silicon for accelerated coverage closure. Highly experienced in full-lifecycle Integration, Verification & Validation (IV&V), IBM DOORS requirement traceability, and multidisciplinary defect triage.

TECHNICAL COMPETENCIES MATRIX

Core Engineering Tooling

FPGA & Digital Logic Silicon Verilog HDL, IEEE VHDL, AMD Xilinx Vivado, Intel Quartus, Artix-7, Zynq-7000, ModelSim RTL, Timing Closure (XDC Constraints), STA, FSM Architecture.	HIL & Test Automation Automation Python HIL Test Engines, PySerial, Automated Test Equipment (ATE), FAT/SAT/ATP Protocols, Signal Injection Rigs, Real-Time Telemetry & Log Triage.
Electro-Optics & Systems Optronics Hawkeye Surveillance Directors, Daylight HD Optics, Cooled Thermal Imagers, Eye-Safe Laser Rangefinders, Optical Collimation, ESS Vibration & Thermal.	Protocols & Instruments Hardware Digital Storage Oscilloscopes, Logic Analysers, Spectrum Analysers, JTAG Boundary Scan, RS-422, CAN Bus, Ethernet, SPI, I2C, UART, PCIe, Camera Link.

PROFESSIONAL WORK EXPERIENCE

Defence Industry

Hardware Test Engineer & Land IPT Test Point of Contact Current Role June 2026 - Present Full-Time Industry Chess Dynamics • Horsham, West Sussex, UK Specialist defence and security enterprise manufacturing high-precision electro-optical surveillance, tracking directors, and stabilised gimbals for international allied defence programmes. Land IPT Test Point of Contact: Serve as the primary test department point of contact for the Land Integrated Portfolio Team (IPT), advising project managers and systems leads on test scoping, verification feasibility, and qualification schedules. Factory Acceptance Testing (FAT): Coordinate and execute formal Factory Acceptance Testing on Hawkeye electro-optical directors, verifying daylight HD optics, cooled continuous-zoom thermal cameras, and eye-safe laser rangefinders against military customer specifications. Hardware Diagnostics & Fault Isolation: Conduct board-level fault diagnostics and root-cause isolation across multi-layer PCBs, motor control servo feedback loops, slip-ring comms (RS-422, CAN, Ethernet), and high-bandwidth video processing pipelines. Optical Boresighting & Calibration: Perform precision optical boresighting and laser rangefinder alignment routines using optical collimators and blackbody emitters to achieve exact weapon-system integration. Environmental Qualification (ESS): Coordinate environmental stress screening, including climatic chamber thermal profiling (-40°C to +70°C) and multi-axis vibration testing to military standards. Documentation & Procedures: Author formal Factory Acceptance Test reports, engineering discrepancy logs, and Standard Operating Procedures (SOPs). Land IPT Coordination Factory Acceptance Testing Optical Boresighting RS-422 / CAN / Ethernet Thermal & Vibration ESS
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Systems Engineer — Integration, Verification & Validation (IV&V) Placement

12-Month Placement

June 2024 - June 2025

BAE Systems • Rochester, Kent, UK

Global tier-one defence and aerospace enterprise delivering mission-critical electronic systems and active vehicle electronics.

- **IV&V Lifecycle Execution:** Executed end-to-end Integration, Verification & Validation (IV&V) procedures on safety-critical electronic systems hardware and testbench architectures.
- **Python Test Automation:** Developed modular Python-based automated test scripts for lab testbenches and signal injection hardware, significantly accelerating regression testing throughput.
- **Requirements Traceability (IBM DOORS):** Managed bidirectional requirements verification matrices in IBM DOORS, mapping high-level system requirements to granular hardware test cases with 100% audit traceability.
- **Signal Integrity & Simulation:** Verified electronic hardware subsystems using signal injection, high-bandwidth oscilloscopes, and MATLAB/Simulink models, including Frequency Response Analysis (FRA).
- **Anomaly Debugging & ERB Reviews:** Collaborated with multidisciplinary hardware, software, and systems teams to diagnose hardware root-cause anomalies, authoring defect reports presented at Engineering Review Boards.
- **STEM Ambassador:** Delivered interactive "STEM in a Box" engineering workshops to over 400 school students across Kent as a certified STEM Ambassador.

Python Automation

IBM DOORS

IV&V Lifecycle

Frequency Response Analysis

Signal Injection

VAIDAR: AI-Driven Hardware-in-the-Loop FPGA Verification Engine

Flagship Project

AMD Xilinx Artix-7

Final Year Capstone Project • First Class 91% (A+) • The IET Prize Winner 2026

Architected an automated hardware-in-the-loop verification engine bridging host software directly to physical FPGA silicon over high-speed 921,600-baud UART to eliminate the 70% lifecycle bottleneck of software simulation.

Physical Hardware Testbed

Real-time vector streaming between host Python runtime and synthesised 16-bit ALU on physical Digilent Nexys A7 FPGA silicon.

Machine Learning Stimulus

Integrated adaptive stimulus generation and automated LLM discrepancy triage, delivering 99.9%+ logic coverage closure.

- Achieved **968.9 tests/second** sustained throughput on physical silicon (~44,000x faster than manual bench testing with 100% physical accuracy).

EDUCATION

BEng (Hons) Electronic & Computer Engineering
University of Brighton • First Class 80%

Graduated July 2026

Level 6: 80% • Level 5: 77% • Level 4: 78%

Specialisms: Digital RTL, Embedded C, HIL Automation, DSP, Closed-Loop Control.

OFFICIAL HONOURS

The IET Prize 2026

Awarded by The Institution of Engineering and Technology for outstanding academic distinction (First Class 80%).

IET Sussex Prize 2023

Awarded by The IET for outstanding first-year academic performance.

COMMERCIAL LEADERSHIP & OPERATIONS

Brighton Real Ale Society (Oct 2025 – June 2026)

Co-President, Vice President & Web Architect

Elected executive leader managing society administration, event licensing, budgeting, and engineered the full-stack web platform using Next.js, Prisma, and Supabase.

bp / M&S Simply Food (Dec 2023 – June 2026)

Customer Service Assistant

Forecourt safety compliance, hazardous materials handling, fresh food temperature audits, and inventory auditing.

McDonald's (Aug 2020 – Mar 2024)

Shift Manager — Eastleigh, Hampshire

Led operations directing shifts of up to 30 personnel under intense tempo, shift scheduling, safe drops, cash reconciliation, and hygiene auditing.

Deliveroo & Jamie's Computers

Urban Courier & PC Refurbisher

Time-critical urban food delivery (Sept 2023 – June 2024); diagnostic testing and hardware refurbishing for low-income families (2021).